Data Networks

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1 M/M/m : Erlang C

1.1

: λ, μ, m

- : $a = \lambda/\mu = m\rho$
- : $\rho = \lambda/(m\mu) < 1$ ( )

CTMC balance equation :

$$p_n = \begin{cases} \dfrac{a^n}{n!}\, p_0 & 0 \le n < m \\ \dfrac{a^n}{m!\, m^{n-m}}\, p_0 & n \ge m \end{cases}$$

1.2 Erlang C

$n \ge m$ $n = m + k$ $(k \ge 0)$:

$$p_{m+k} = \frac{a^{m+k}}{m!\, m^k}\, p_0 = \frac{a^m}{m!} \cdot \rho^k \cdot p_0 = p_0 \cdot \frac{(m\rho)^m}{m!} \cdot \rho^k$$

busy :

$$P_Q = \sum_{n=m}^{\infty} p_n = p_0 \cdot \frac{(m\rho)^m}{m!} \sum_{k=0}^{\infty} \rho^k$$

$\sum_{k=0}^{\infty} \rho^k = 1/(1-\rho)$:

$$\boxed{P_Q = \frac{p_0\,(m\rho)^m}{m!\,(1-\rho)}} \quad \text{(Erlang C)}$$

1.3 N_Q

! : k $m+k$?

N_Q . n m $(n-m)$.

$$N_Q = \sum_{n=m}^{\infty} (n-m)\,p_n$$

$n = m+k$ $(n-m) = k$:

$$N_Q = \sum_{k=0}^{\infty} k \cdot p_{m+k}$$

$(m+k)$ $E[N]$.

— ρ :

$$\sum_{k=0}^{\infty} k\,\rho^k = \rho \cdot \frac{d}{d\rho} \sum_{k=0}^{\infty} \rho^k = \frac{\rho}{(1-\rho)^2}$$

:

$$N_Q = p_0 \cdot \frac{(m\rho)^m}{m!} \cdot \frac{\rho}{(1-\rho)^2} = \boxed{P_Q \cdot \frac{\rho}{1-\rho}}$$

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2 M/M/∞:

2.1 Detailed Balance

n $n\mu$:

$$\lambda\,p_{n-1} = n\mu\,p_n \quad \Rightarrow \quad p_n = p_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!}$$

2.2 $p_0 = e^{-\lambda/\mu}$

$\sum_{n=0}^{\infty} p_n = 1$:

$$p_0 \sum_{n=0}^{\infty} \frac{(\lambda/\mu)^n}{n!} = 1$$

:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \;\Rightarrow\; \sum_{n=0}^{\infty} \frac{(\lambda/\mu)^n}{n!} = e^{\lambda/\mu}$$

:

$$\boxed{p_0 = e^{-\lambda/\mu}}$$

2.3 : Poisson

$$p_n = \frac{(\lambda/\mu)^n \, e^{-\lambda/\mu}}{n!} \sim \text{Poisson}(\lambda/\mu)$$

> M/M/∞ . Exp(μ) . Poisson + (ρ), Poisson .

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3 M/M/m/m: (Kendall)

3.1

$M/M/m/K$

1	(M = Poisson)
2	(M = Exponential)
3	
4	

M/M/m/m . $\{0, 1, \ldots, m\}$, m .

3.2 Erlang B ( )

$$p_n = \frac{(\lambda/\mu)^n/n!}{\sum_{k=0}^{m} (\lambda/\mu)^k/k!}$$

$$\boxed{P_{\text{loss}} = p_m = \frac{(\lambda/\mu)^m/m!}{\sum_{k=0}^{m} (\lambda/\mu)^k/k!}}$$

Circuit switching **(blocking probability)** .

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4 M/G/1 : Pollaczek-Khinchine

4.1 P-K

$$W = \frac{\lambda E[X^2]}{2(1-\rho)}, \qquad N_Q = \frac{\lambda^2 E[X^2]}{2(1-\rho)}$$

4.2 : 3

P-K + **PASTA** + **Little's Law** .

4.2.1 Step 1:

W:

$$W = \underbrace{R} + \underbrace{\sum_{i=1}^{N_Q} X_i}$$

PASTA(Poisson Arrivals See Time Averages) $N_Q =$ N_Q.

:

$$W = E[R] + N_Q \cdot E[X]$$

4.2.2 Step 2: $E[R]$

busy ρ :

$$E[R] = \rho \cdot E[\quad \mid \text{busy}]$$

Inspection Paradox (Size-biased):

x — :

$$f_{\tilde{X}}(x) = \frac{x \cdot f_X(x)}{E[X]}$$

x $[0, x]$:

$$E[\quad \mid X = x] = \frac{x}{2}$$

:

$$E[\quad \mid \text{busy}] = \int_0^\infty \frac{x}{2} \cdot \frac{x f_X(x)}{E[X]} \, dx = \frac{E[X^2]}{2E[X]}$$

$\rho = \lambda E[X]$:

$$E[R] = \rho \cdot \frac{E[X^2]}{2E[X]} = \frac{\lambda E[X^2]}{2}$$

4.2.3 Step 3: Little's Law

$N_Q = \lambda W$ Step 1 :

$$W = \frac{\lambda E[X^2]}{2} + \lambda W \cdot E[X] = \frac{\lambda E[X^2]}{2} + \rho W$$

W :

$$\boxed{W = \frac{\lambda E[X^2]}{2(1-\rho)}}$$

4.3

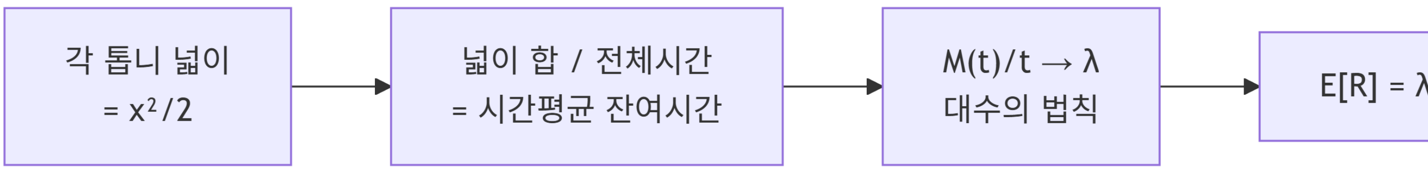
$r(\tau)$, :

- ↑: , x_i
- : -1
- 0 :

i $= \frac{1}{2}x_i^2$

:

$$\frac{1}{t}\int_0^t r(\tau)\,d\tau = \frac{\sum_i \frac{1}{2}x_i^2}{t} \xrightarrow{t\to\infty} \lambda \cdot \frac{E[X^2]}{2}$$



$E[X^2]$.

4.4

M/M/1 $E[R]$?

:

$$P(X > s+t \mid X > s) = P(X > t)$$

. M/M/1 :

$$E[\quad \mid \text{busy}] = E[X] = 1/\mu$$

. M/G/1 Inspection Paradox $E[R]$, $E[X^2]$.

4.5 $E[X^2] = 2/\mu^2$

:

$$E[X^2] = \mathrm{Var}(X) + (E[X])^2 = \frac{1}{\mu^2} + \frac{1}{\mu^2} = \frac{2}{\mu^2}$$

(2):

$$E[X^2] = \int_0^\infty x^2 \mu e^{-\mu x}\,dx = \frac{2}{\mu^2}$$

4.6 M/M/1 vs M/D/1

	M/M/1	M/D/1 (Deterministic)
	Exp(μ)	$1/\mu$
Var(X)	$1/\mu^2$	**0**
$E[X^2]$	$2/\mu^2$	$1/\mu^2$
W	$\dfrac{\rho}{\mu(1-\rho)}$	$\dfrac{\rho}{2\mu(1-\rho)}$

! P-K

$$W \propto E[X^2] = \mathrm{Var}(X) + (E[X])^2$$

. D (=0) , M 2 .

4.7 : Go-Back-n ARQ

p: , n: . X ( ):

$$P[X = 1 + kn] = (1-p)p^k, \quad k = 0, 1, \ldots$$

$K \sim \text{Geometric}(1-p)$:

- $E[K] = \dfrac{p}{1-p}$
- $E[K^2] = \dfrac{p(1+p)}{(1-p)^2}$

$E[X]$ **:**

$$E[X] = 1 + n \cdot E[K] = 1 + \frac{np}{1-p}$$

$E[X^2]$ ($X = 1 + Kn$):

$$E[X^2] = 1 + 2nE[K] + n^2 E[K^2] = 1 + \frac{2np}{1-p} + \frac{n^2(p+p^2)}{(1-p)^2}$$

P-K ARQ .

: $p \uparrow \rightarrow E[X^2]$ $(1-p)^2$ $\rightarrow$.

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5 Reservations and Polling

5.1

Statistical multiplexing , :

$$\text{Cycle} = \text{Reservation interval} + \text{Data interval}$$

5.2

Fully Gated	Reservation
Partially Gated	Data
Exhaustive	( )

5.3 User 1 Arrival Interval

“Transmission interval for user 1” :

5.3.1 Fully Gated

User 1 **Reservation** ~ User 1 Reservation → **Full cycle**

5.3.2 Partially Gated

User 1 **Data** ~ User 1 Data → Full cycle − User 1 Reservation

5.3.3 Exhaustive

User 1 **Data** ~ User 1 Data → Full cycle − (User 1 Reservation + Data)

5.4 Arrival Interval

L $L/2$:

$$W_{\text{Fully Gated}} > W_{\text{Partially Gated}} > W_{\text{Exhaustive}}$$

Exhaustive Reservation + Data Arrival Interval .

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6 Tandem Queues Burke’s Theorem

6.1

```
Poisson( ) → | Queue 1 | → | Queue 2 | → ...
```

Queue 1 departure Queue 2 arrival . Queue 2 departure .

6.2 M/M/1 Departure

! Burke’s Theorem

M/M/1 (M/M/m, M/M/∞) :

1. **Departure** **rate λ Poisson process**
2. t t **departure**

→ Queue 2 **M/M/1**

6.2.1 Burke's Theorem : Time Reversibility

6.2.1.1 Backward Transition Probability

+ Markov :

$$P[X_m = j \mid X_{m+1} = i, X_{m+2} = i_2, ...] = \frac{p_j P_{ji}}{p_i}$$

$m+2$, Markov .

6.2.1.2 Time Reversible

$$P^*_{ij} = P_{ij} \quad \forall i, j$$

Detailed Balance Equation:

$$\boxed{p_i P_{ij} = p_j P_{ji}}$$

6.2.1.3 M/M/1 Time Reversible

Birth-Death detailed balance $p_n \lambda = p_{n+1}(n+1)\mu$.

6.2.1.4 Time Reversibility Burke

- M/M/1
- " " = " "
- Poisson(λ) $\rightarrow$ **departure Poisson(λ)**

6.3 M/D/1 M/G/1 Departure

6.3.1 M/D/1 (= d)

$$\text{inter-departure} = \max(\text{inter-arrival}, d)$$

d .

6.3.2 M/G/1 ( )

X_i ( ) inter-departure time :

- X_i $\rightarrow$ Queue 1 $\rightarrow$ $\rightarrow$ departure
- X_i $\rightarrow$ $\rightarrow$ idle $\rightarrow$ departure

$\rightarrow$ Queue 2 Poisson Renewal .

6.4 Tandem

Queue 1	Departure	Queue 2	
M/M/1	Poisson	**M/M/1**	(Burke)
M/D/1 ( )	Deterministic	**= 0** ( )	
M/D/1 ( )			
M/G/1			

6.4.1 M/D/1

Queue 1 Queue 2 d :

$$D_i = \max(A_i, d) \geq d = \text{Queue 2}$$

Queue 2 **Queue 2 = 0**.

7 Kleinrock Independence Assumption

7.1

M/G/1 Tandem → .

7.2

( ) inter-arrival time . → **M/M/1** .

(mix) .

7.3 (i,j)

$$N_{ij} = \frac{\lambda_{ij}}{\mu_{ij} - \lambda_{ij}}$$

7.4

$$N = \sum_{(i,j)} \frac{\lambda_{ij}}{\mu_{ij} - \lambda_{ij}}$$

Little's Law $N = \gamma T$:

$$T = \frac{1}{\gamma} \sum_{(i,j)} \frac{\lambda_{ij}}{\mu_{ij} - \lambda_{ij}}$$

$\gamma = \sum_p x_p$.

7.5 p

3 :

$$T_p = \sum_{(i,j) \in p} \left(\underbrace{\frac{\lambda_{ij}}{\mu_{ij}(\mu_{ij} - \lambda_{ij})}} + \underbrace{\frac{1}{\mu_{ij}}} + \underbrace{d_{ij}}_{+} \right)$$

7.6

7.6.1 Example 1: Randomization (M/M/1)

$$T_R = \frac{1}{\mu - \lambda/2} = \frac{2}{2\mu - \lambda}$$

7.6.2 Example 2: Metering (M/M/2)

$$T_M = \frac{1}{\mu} + \frac{P_Q}{2\mu - \lambda}$$

M/M/2 Erlang C :

$$T_M = \frac{2}{2\mu - \lambda} \cdot \frac{1}{1+\rho}$$

7.6.3

$$\frac{T_M}{T_R} = \frac{1}{1+\rho} < 1$$

M/M/2 Metering , ρ . (pooling) .

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8 : $E[E[X|Y]] = E[X]$

8.1 Tower Property (Law of Total Expectation)

X, Y :

$$\boxed{E[E[X \mid Y]] = E[X]}$$

Tower Property **(Law of Total Expectation)** .

8.2

:

$$E[X \mid Y = y] = \sum_x x \cdot P(X = x \mid Y = y)$$

Y :

$$E[E[X \mid Y]] = \sum_y E[X \mid Y = y] \cdot P(Y = y)$$

:

$$= \sum_y \left[\sum_x x \cdot P(X = x \mid Y = y) \right] P(Y = y)$$

$P(X = x \mid Y = y)P(Y = y) = P(X = x, Y = y)$:

$$= \sum_y \sum_x x \cdot P(X = x, Y = y)$$

(Fubini):

$$= \sum_x x \sum_y P(X = x, Y = y)$$

: $\sum_y P(X = x, Y = y) = P(X = x)$

$$= \sum_x x \cdot P(X = x) = E[X] \quad \blacksquare$$

8.3

:

$$E[E[X \mid Y]] = \int_{-\infty}^{\infty} E[X \mid Y = y] f_Y(y)\, dy$$

$$= \int_y \left[\int_x x \cdot f_{X|Y}(x|y)\, dx \right] f_Y(y)\, dy$$

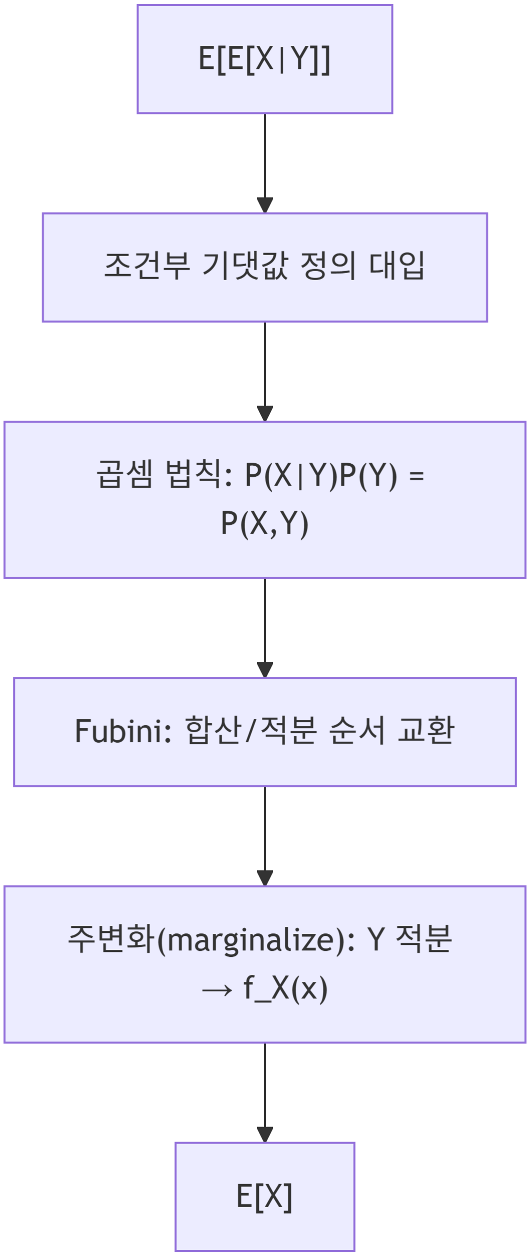
$f_{X|Y}(x|y) f_Y(y) = f_{X,Y}(x, y)$:

$$= \int_y \int_x x \cdot f_{X,Y}(x, y)\, dx\, dy$$

Fubini :

$$= \int_x x \underbrace{\int_y f_{X,Y}(x, y)\, dy}_{= f_X(x)} dx = \int_x x \cdot f_X(x)\, dx = E[X] \quad \blacksquare$$

8.4



8.5

$E[X \mid Y]$ Y . Y X , Y $E[X]$.

8.6

Tower Property :

- : $E[R] = E[E[R \mid X]]$
- **Go-Back-n ARQ**: $E[X] = E[E[X \mid K]]$ ($K =$)
- **Jackson Network**: marginalize
- :

8.7 :

Tower Property **Law of Total Variance** :

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y])$$

" ", " " .

9

9.1

	p_n	N N_Q	
M/M/1	$(1-\rho)\rho^n$	$N = \dfrac{\rho}{1-\rho}$	
M/M/m		$N_Q = P_Q \cdot \dfrac{\rho}{1-\rho}$	Erlang C
M/M/∞	$\dfrac{(\lambda/\mu)^n e^{-\lambda/\mu}}{n!}$	$N = \lambda/\mu$	Poisson
M/M/m/m		$P_{\text{loss}} = p_m$	Erlang B
M/G/1		$N_Q = \dfrac{\lambda^2 E[X^2]}{2(1-\rho)}$	P-K

9.2

Little's Law $N = \lambda T$	/
PASTA	Poisson =
Burke's Theorem	M/M/1 departure = Poisson
Kleinrock Independence	M/M/1
Tower Property	$E[X] = E[E[X \mid Y]]$
Detailed Balance	$p_i P_{ij} = p_j P_{ji}$ (reversibility)

9.3

ℹ

1. $n = m + k$: M/M/m $n \geq m$
2. : $\sum k\rho^k = \rho/(1-\rho)^2$
3. : $\sum (x^n/n!) = e^x$
4. **Inspection Paradox**: size-biased
5. **Tower Property**:
6. **Little's Law** : $N_Q = \lambda W$

10

1. **Erlang C B** : (C) vs (B)
2. $E[X^2]$:
3. **M/M/1 vs M/G/1**
4. **Burke's Theorem M/M**
5. **M/D/1 Tandem** ( )
6. **Kleinrock Independence**
7. **Reservation Arrival Interval**
8. **Tower Property**